BADER ALE

AI Engineer | Generative AI, LLM Applications, RAG, Agentic AI & MLOps

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PROFESSIONAL SUMMARY

AI Engineer who designs, builds, and deploys production generative AI systems end to end: Retrieval-Augmented Generation (RAG) pipelines, multi-agent LLM workflows, Model Context Protocol (MCP) tool servers, and machine learning models. Founder of Taxentia.ai, a live RAG SaaS for tax-law research. Ships in Python, FastAPI, LangChain/LangGraph, OpenAI and Anthropic Claude APIs, and vector databases, deployed with Docker, GitHub Actions CI/CD, and AWS. Owns the full lifecycle from data ingestion and prompt/context engineering to deployment, guardrails, and iteration. Engineering foundation (B.S. Mechanical Engineering, M.S. Engineering Management) plus graduate data analytics training. Builds for regulated industries (legal, finance, healthcare) with HIPAA-aware design. Bilingual English/Spanish.

CORE COMPETENCIES

Generative AI • Large Language Models (LLMs) • Retrieval-Augmented Generation (RAG) • Agentic AI & Multi-Agent Orchestration • Model Context Protocol (MCP) • Prompt & Context Engineering • Embeddings & Semantic Search • Vector Databases • AI Guardrails & Model Evaluation • MLOps / LLMOps • Machine Learning • Natural Language Processing (NLP) • Data Pipelines (ETL/ELT) • API Development • Cloud Deployment • CI/CD

TECHNICAL SKILLS

LLM & Agentic AI: OpenAI API, Anthropic Claude API, LangChain, LangGraph, RAG, MCP servers, AI agents and copilots, multi-agent systems, tool/function calling, prompt engineering, embeddings, Pinecone, vector databases, Claude Code

Machine Learning: PyTorch, TensorFlow, scikit-learn, XGBoost, Gradient Boosting, Random Forest, Logistic Regression, Markov models, feature engineering, model evaluation (ROC/AUC), Jupyter

Languages: Python, SQL, TypeScript, R

Backend & Data: FastAPI, REST APIs, Supabase/PostgreSQL, Apache Spark, ETL/ELT, NATS messaging, event-driven microservices, Tableau, Power BI

Cloud & DevOps: AWS, Docker, GitHub Actions, CI/CD, Git, Linux, uv

Automation & Integrations: n8n, Slack API, WhatsApp API, CRM integrations (GoHighLevel), webhooks

Compliance & Quality: HIPAA-aware architecture, HL7/FHIR, Lean Six Sigma

PROFESSIONAL EXPERIENCE

AI Engineer & Founder | Taxentia.ai and Independent AI Engineering (Remote) **2016 – Present**

- Architect and ship end-to-end **generative AI** systems (RAG pipelines, LLM agents, MCP integrations, ML models) across legal, finance, healthcare, and real-estate use cases
- Own the full AI lifecycle: data ingestion and ETL, chunking and embeddings, retrieval design, prompt and context engineering, **FastAPI** service layer, containerized deployment (**Docker, GitHub Actions CI/CD, AWS**), and post-launch iteration
- Built **agentic workflow automations** connecting LLMs to Slack, WhatsApp, CRMs, and n8n, replacing manual business processes; cut realtor follow-up workload **30–40%**
- Translate business requirements into AI solutions; select models and architectures based on accuracy, latency, cost, and compliance tradeoffs

Mechanical / Project Engineer | M2E, AMETEK (FAA), GA Telesis, Bouygues **2009 – 2015**

- Mechanical, MEP, and aerospace tooling engineering in regulated, safety-critical environments; **~20%** energy-efficiency gains and FAA-compliant tooling with **~15%** less downtime (SolidWorks, Six Sigma)
- Network / Technical Support Engineer, FIU College of Engineering & Computing: **30%** less downtime, **40%** fewer support tickets

AI PROJECTS

Taxentia.ai – Production RAG SaaS (taxentia.ai) | *Python, OpenAI, Pinecone, Supabase, Docker, GitHub Actions*

- Built and launched a production **RAG** application for real-time tax-law retrieval and automated memo generation
- Engineered document ingestion pipeline (parsing, chunking, embeddings, vector indexing) and Supabase/PostgreSQL schema with tiered authentication and PDF export

Axon-Health – Federated Healthcare AI Platform | *Python, NATS, microservices*

- Architected an event-driven, federated platform of modular Python services on a NATS message bus for compliant deployment of clinical AI services
- Designed two-tier HIPAA and compliance gatekeepers acting as **AI guardrails** on data flowing between services

Monarch-MCP – Model Context Protocol Server | *Python, MCP, Claude*

- Built an **MCP server** giving LLM agents secure tool access to financial account and credit card data for natural-language financial analysis

LeadWhisperer.ai – Conversational AI Assistant | *OpenAI, Slack/WhatsApp APIs, Supabase CRM*

- Built an LLM-powered MLS assistant for realtors with automated follow-ups, reducing manual workflow **30–40%**

APEX – ML Trading Signal Engine | *Python, scikit-learn, Jupyter*

- Built a Markov regime detector, pattern scanner, and **Gradient Boosting** signal engine for paper-trading SPY 0DTE and pharma-catalyst setups

Heart Failure Prediction – Clinical Machine Learning | *Python, scikit-learn, XGBoost*

- Built classification models (Logistic Regression, Random Forest, XGBoost) with ETL and feature engineering; achieved **85%+ accuracy** validated with ROC/AUC

EDUCATION

M.S., Data Analytics (in progress) – Western Governors University

M.S., Engineering Management – Florida International University

B.S., Mechanical Engineering – Florida International University

Professional Certificate, Robotics Engineering – Florida International University

CERTIFICATIONS

AWS Cloud Quest: Cloud Practitioner • Engineer Intern (EIT), Florida Board of Professional Engineers • Six Sigma Green Belt • Data Preparation Micro-credential (WGU) • CompTIA A+ • CompTIA Network+ • Microsoft Certified Professional

LANGUAGES

English (fluent) • Spanish (fluent) • Italian (basic)